2 . Multilayer Perceptron

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, log_loss

iris = load_iris()
X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

models = {
    "1 Hidden Layer - ReLU": MLPClassifier(
        hidden_layer_sizes=(10,),
        activation="relu",
        solver="adam",
        max_iter=1000,
        random_state=42
    ),
    "2 Hidden Layers - ReLU": MLPClassifier(
        hidden_layer_sizes=(10, 10),
        activation="relu",
        solver="adam",
        max_iter=1000,
        random_state=42
    ),
    "2 Hidden Layers - Tanh": MLPClassifier(
        hidden_layer_sizes=(10, 10),
        activation="tanh",
        solver="adam",
        max_iter=1000,
        random_state=42
    )
}
```

```
results = []

for name, model in models.items():
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)
    y_prob = model.predict_proba(X_test)

    accuracy = accuracy_score(y_test, y_pred)
    loss = log_loss(y_test, y_prob)

    results.append([name, accuracy, loss])

    print(name)
    print("Accuracy:", accuracy)
    print("Loss:", loss)
    print()

names = [r[0] for r in results]
accuracies = [r[1] for r in results]
losses = [r[2] for r in results]

x = np.arange(len(names))

plt.figure(figsize=(10, 5))
plt.bar(x, accuracies)
plt.xticks(x, names, rotation=20)
plt.ylabel("Accuracy")
plt.title("MLP Accuracy Comparison")
plt.ylim(0, 1.1)
plt.show()

plt.figure(figsize=(10, 5))
plt.bar(x, losses)
plt.xticks(x, names, rotation=20)
plt.ylabel("Loss")
plt.title("MLP Loss Comparison")
plt.show()
```

```
1 Hidden Layer - ReLU
Accuracy: 0.9666666666666667
Loss: 0.10790371977761341

2 Hidden Layers - ReLU
Accuracy: 0.9666666666666667
Loss: 0.08491795104929155

2 Hidden Layers - Tanh
Accuracy: 0.9666666666666667
Loss: 0.08485333647090833
```

